COMPUTATIONAL ELEMENTS OF DATA GOVERNANCE

The term *federated computational governance* was coined by Dehghani in her de facto data mesh manifesto (see her blog article, "Data Mesh Principles and Logistical Architecture," http://mng.bz/44rV). However, as computational governance is not defined anywhere else, throughout this book we will assume that this term relates to elements of data governance automation enabled through the existence of the central platform (as we call self-serve infrastructure as a platform, presented in a bit more detail in section 1.4.4 and in a whole lot more detail in chapters 7 and 8) serving as a medium connecting different data products.

Data governance elements, which can be automated and embedded into a data platform, include but are not limited to the following:

- Metadata
- Catalog, reference, and master data
- Lineage and provenance
- Validation and verification
- Storage and operations
- Security and privacy

Once again, no silver-bullet solution exists for deciding which of these should be automated. It will require your teams' hard work to identify which data governance tasks create bottlenecks in data mesh development in your organization and to automate them. However, automation will pay off. First, it will offer data product owners the solution that allows them a frictionless connection of their data products. Second, it will enable users to make efficient use of the exposed data. You will learn more about automating elements of data governance when we discuss the details of the self-serve data infrastructure as a platform next.

1.4.4 Self-serve data infrastructure as a platform

The fourth principle is to *extract the duplicated efforts of the data mesh into a platform*. This principle calls for the application of platform thinking to the